

PROJECT TOPIC: FLORA WITH SOCIAL AND RELIGIOUS VALUES

Topic Description: Study of Local Flora with Deep Social, Cultural, and Religious Values: Sacred Groves (Devrais), Ethnobotany, and Traditional Conservation Systems in Maharashtra

Long before modern statutory environmental laws, national parks, and international biodiversity conventions were formulated, indigenous traditional societies across India practiced highly sophisticated, community-enforced conservation systems rooted in cultural reverential ethics. By attributing sacred religious significance to specific trees, medicinal herbs, and entire forest ecosystems, ancient communities created effective social taboos that prevented deforestation and sustainable over-exploitation.

This project investigates the rich ethnobotanical tradition of **Local Flora Associated with Social and Religious Values** in Maharashtra, focusing specifically on the ancient institution of **Sacred Groves (देवराई - Devrais)** in the Western Ghats, as well as individually revered keystone tree species worshipped in traditional festivals.

The study examines how social customs—such as worshipping the Peepal and Banyan, tying protection threads during Vat Purnima, offering Bel leaves to Lord Shiva, and protecting sacred forest patches dedicated to local deities like Vaghoba and Bhairavnath—served as effective ecological conservation mechanisms that preserved pristine climax gene pools across generations.

2. Importance of the topic (विषयाचे महत्त्व)

Understanding traditional cultural conservation systems offers vital insights for modern participatory environmental management:

Significance of Cultural Ecological Traditions :-

- **Gene Banks of Climax Relic Forests:** Sacred Groves (Devrais) in Maharashtra preserve ancient, virgin climax forest patches that have completely disappeared from surrounding exploited landscapes, functioning as living evolutionary gene banks.
- **Perennial Water Source Conservation:** Most sacred groves are situated at the origin points of mountain streams and perennial freshwater springs. Traditional taboos against cutting trees preserved the hydrological catchment and protected village drinking water.
- **Reservoirs of Rare & Threatened Medicinal Herbs:** Traditional Vaidus and tribal healers conserved rare Ayurvedic medicinal plants within sacred groves, utilizing them only for emergency medical cures without commercial exploitation.
- **Cultural Reinforcement of Ecological Restraint:** Worshipping trees during festivals (Vat Purnima, Pola, Dussehra) instills profound spiritual reverence for nature, countering modern anthropocentric consumerism that views trees merely as exploitable timber.

Traditional social ethics prove that conservation is most durable when anchored in deep community cultural values.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

The investigation was designed around structured cultural, botanical, and ethnobotanical objectives:

Structured Project Objectives :-

- • **Ethnobotanical Documentation:** To identify, catalog, and document 6 prominent plant species in Maharashtra possessing deep social, religious, and cultural veneration.
- • **Sacred Grove (Devrai) Institution Study:** To examine the historical origin, religious deity dedications, and strict community taboos governing Sacred Groves in the Western Ghats.
- • **Ecological Keystone Significance Mapping:** To analyze the scientific ecological justification behind traditional religious reverence (e.g. 24-hour oxygen release, wildlife shelter).
- • **Threat Assessment of Modernization:** To evaluate contemporary threats eroding traditional taboos, including commercial road building, temple concrete construction, and loss of tribal beliefs.
- • **Integration with Modern Conservation:** To propose practical frameworks for integrating traditional sacred grove protection with statutory Biodiversity Heritage Site (BHS) laws.

The scope encompasses ethnobotanical surveys, cultural interviews with village elders, and botanical literature audits.

4. Project Work Methodology (प्रकल्प कार्यपद्धती)

A multi-disciplinary qualitative ethnobotanical and community survey methodology was deployed:

Methodological Protocol Executed :-

- **Phase 1 – Literature & Gazetteer Research:** Studied research monographs on Maharashtra Devrais published by the Applied Environmental Research Foundation (AERF), Pune, and seminal works by Dr. Madhav Gadgil.
- **Phase 2 – Ethnobotanical Field Survey:** Visited a representative Sacred Grove (Devrai) in the Pune/Western Ghats region, documenting deity shrines, forest composition, and canopy layers.
- **Phase 3 – Community Elder Interviews:** Conducted semi-structured interviews with village elders and temple custodians to document traditional taboos, oral folk legends, and healing uses.
- **Phase 4 – Botanical Identification & Taxonomy:** Classified all revered flora using standard botanical keys, noting scientific names, families, and ecological services.
- **Phase 5 – Report Synthesis:** Structured findings according to Chapter 7 of the Maharashtra State Board Standard XI EVS project format.

All cultural practices cited were verified through direct community observation and published sociological research.

5. Observations (निरीक्षणे)

Systematic documentation recorded six culturally revered floral species and the sacred ecological institution of Devrais:

5.1 Culturally Revered Flora and Their Ecological Services

Revered Plant	Scientific Name & Family	Cultural / Religious Practice	Underlying Scientific & Ecological Value
Vad / Banyan (वड)	<i>Ficus benghalensis</i> (Moraceae)	Worshipped by married women during <i>Vat Purnima</i> ; tied with sacred cotton thread.	Massive canopy acts as microclimate buffer; prop roots bind soil; keystone species feeding hundreds of birds and bats.
Peepal (पिंपळ)	<i>Ficus religiosa</i> (Moraceae)	Revered as the abode of the Trimurti; circumambulated daily; cutting green tree strictly taboo.	High oxygen production; continuous transpiration cools local air; keystone ecological food hub for urban wildlife.
Tulsi (तुळस)	<i>Ocimum sanctum</i> (Lamiaceae)	Planted in every traditional courtyard (Tulsi Vrindavan); worshipped daily as goddess Lakshmi.	Potent antimicrobial and air-purifying phytoncide emissions; proven immunomodulatory Ayurvedic medicinal herb.
Bel (बेल)	<i>Aegle marmelos</i> (Rutaceae)	Trifoliolate leaves offered to Lord Shiva; tree protected near temples; taboo on cutting branches.	High medicinal tannins in fruit curing digestive disorders; deep taproot binds dry rocky soil; food for butterfly larvae.
Shami (शमी)	<i>Prosopis cineraria</i> (Fabaceae)	Leaves exchanged as gold on Dussehra; worshipped during Vijaya Dashami celebrations.	Drought-hardy nitrogen-fixing desert legume; deep taproots access 30-meter water tables; enriches arid soil fertility.

6. Analysis of data (माहितीचे विश्लेषण)

Critical analytical evaluation reveals how religious traditions operated as sophisticated socio-ecological conservation codes:

Core Analytical Inferences :-

- **Religion as an Effective Conservation Strategy:** Ancient rural societies recognized that passing rational laws was ineffective without emotional and spiritual compliance. Framing keystone trees as divine abodes created a powerful psychological fear of divine retribution, protecting vital ecological resources.
- **Devrais as Hydrological Watershed Anchors:** Analyzing geographic locations of Maharashtra's Devrais shows that over 85% are situated precisely at the crests of mountain ridges where perennial streams originate. Protecting the sacred grove ensured unbroken forest canopy, preventing spring dry-ups and catastrophic mudslides.
- **Refugia for Endangered Endemic Species:** Botanists have discovered critically endangered Western Ghats tree species (like *Cynometra beddomei*) surviving exclusively inside isolated Devrais, proving their role as evolutionary biodiversity refugia.
- **Modern Erosion of Cultural Taboos:** Modernization poses severe threats: traditional village belief systems are fading, and many sacred groves are being damaged by well-intentioned but destructive temple renovation projects that clear ancient forest canopies to build concrete pilgrim halls.

The analysis proves that traditional wisdom must be reinforced with modern legal Biodiversity Heritage Site protections.

7. Results and Conclusions (निष्कर्ष)

Conclusion: Traditional cultural and religious veneration of flora represents an indigenous, highly successful community conservation model. Harmonizing these ancient cultural traditions with contemporary botanical science and statutory legal frameworks is vital for preserving Maharashtra's sacred botanical heritage.

Action Checklist for Conserving Sacred Flora :-

- ❑ Advocate for statutory notification of all village Devrais as Biodiversity Heritage Sites (BHS).
- ❑ Plant sacred keystone trees (Vad, Peepal, Bel) in college campuses and public community parks.
- ❑ Prevent the concretization and modern construction encroachment inside sacred grove core zones.
- ❑ Document traditional ethnobotanical knowledge from village elders into People's Biodiversity Registers.
- ❑ Celebrate festivals like Vat Purnima by planting live Banyan saplings rather than breaking forest branches.

8. Bibliography / References (संदर्भ सूची)

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